

Vehicle Miles Traveled Taxes Rollout across States

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Vehicle miles traveled ([VMT](#)) taxes are the road funding tool of the future. For decades, the gas tax proved to be a reliable user fee that funded road construction and maintenance, and consumers paid for their road usage via gasoline taxes when they filled up at the pump. Unfortunately, gas tax revenues have decoupled from road expenses and have been unable to [support road funding](#) in recent years. As such, states nationwide are exploring ways to supplement or replace gas tax revenues.

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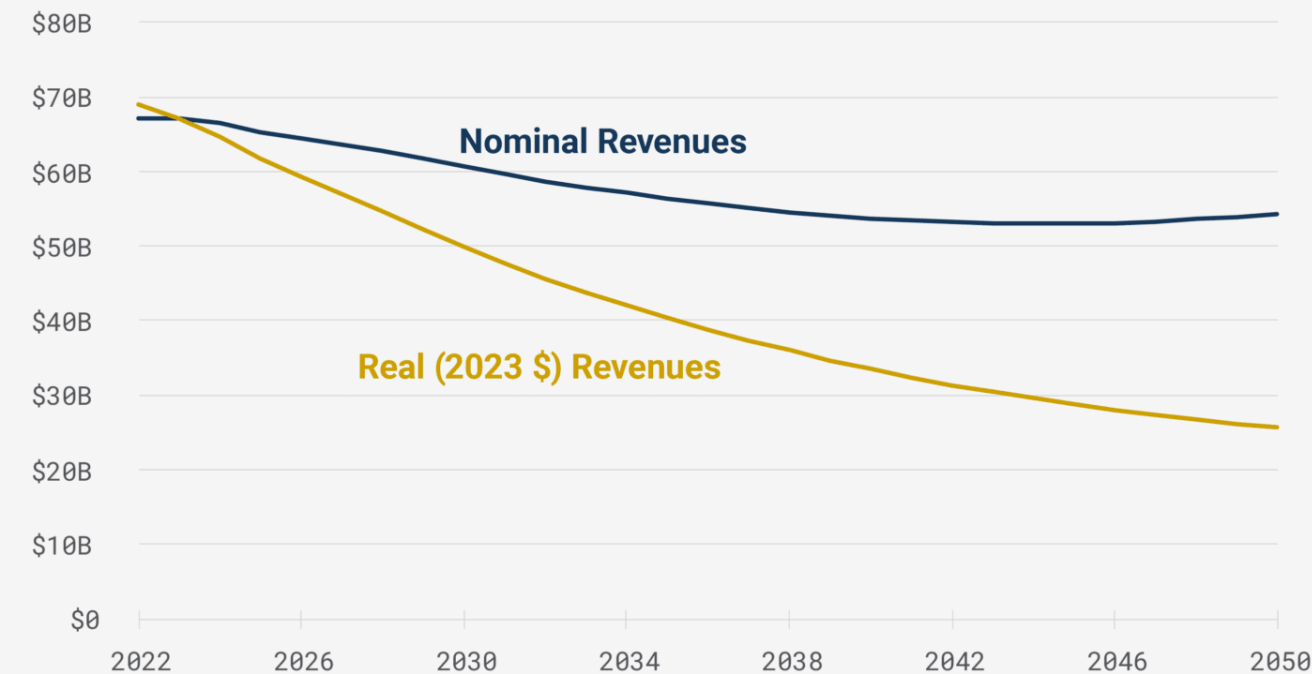
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Decline in Gasoline Consumption Is Eroding Gas Tax Revenue

Gasoline Consumption Decline Will Erode Gas Tax Revenue

Projected Annual State and Federal Gas Tax Revenue Calculated by Projected Gasoline Consumption



Source: U.S. Energy Information Administration, Annual Energy Outlook 2023; Author's calculations.



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In 2023, Americans drove more miles than ever before. However, real revenues from gas taxes have been falling. Figure 1 shows that real gas tax revenues are projected to contract by more than half of their current level over the next 20 years and fall to nearly a

third

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widespread benefits. When combined with gas tax rates that have not kept pace with inflation, however, the result is that gas tax revenues have become increasingly insufficient to cover roadway expenses.

Exploring an Efficient Solution

An efficient solution would be to replace the existing system of gas taxes, registration fees, and other miscellaneous surcharges with a VMT tax. Such taxes are also called mileage-based user fees, road usage charges, or something similar, but they are all user fees that strive to establish a better road funding mechanism by aligning use with costs.

A VMT tax is a charge that drivers pay per mile they drive on public roads, which may vary based on the vehicle's weight per axle and/or the time and location of the driving. Instead of using gas consumption as a proxy for road use, a VMT tax would measure actual road use and then charge the driver for each mile driven. If accurately calibrated, this would reconcile roadway revenues and expenditures, eliminate the redistributive effects of the gas tax, and insulate road funding from changing efficiencies, technologies, or consumer preferences. With deficits only increasing, change is needed sooner rather than later.

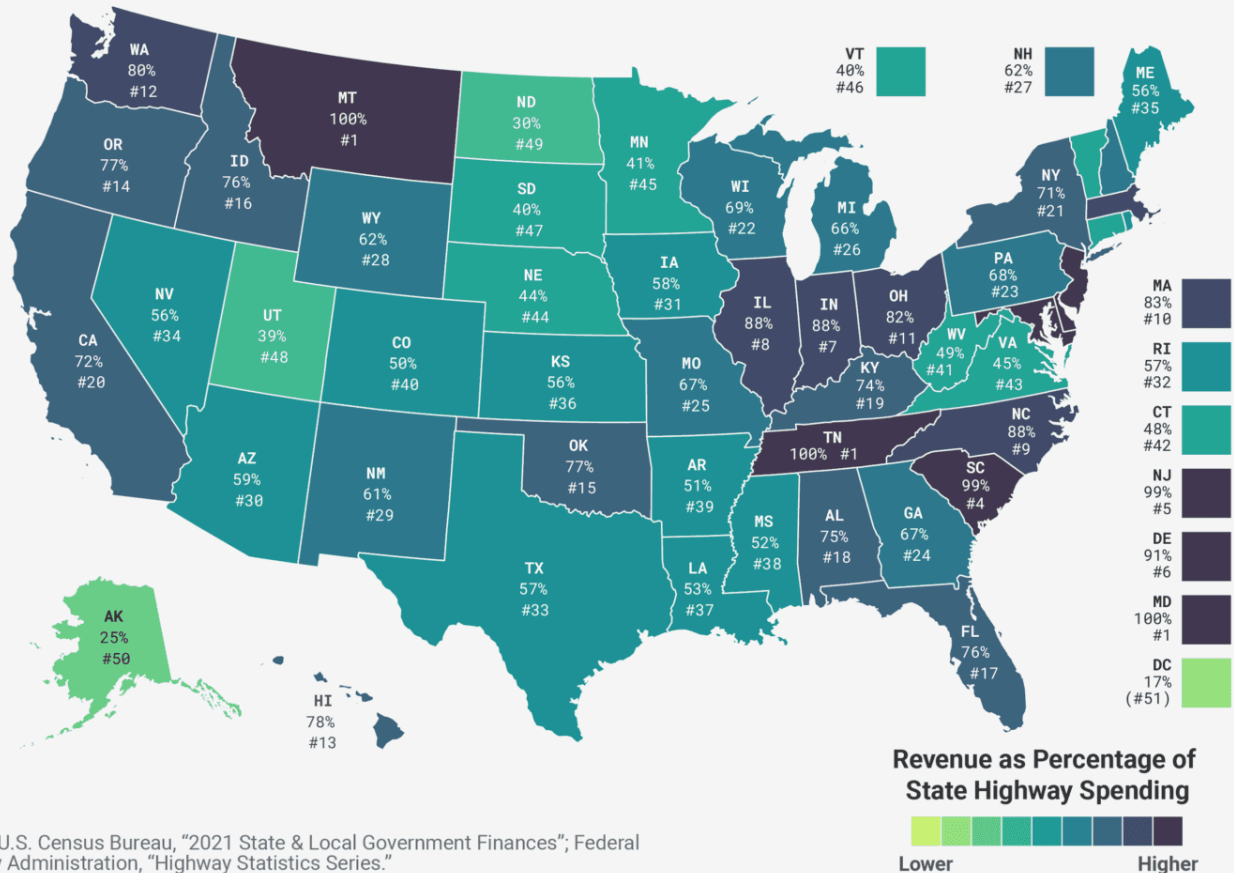
Currently, only three states can fully fund their share of road spending with transportation revenues. In other states, the remaining funds must be diverted from more general revenue sources. This means that non-drivers are forced under the current system to subsidize drivers and pay for roads that they do not use. It also means that other areas of spending must work with less after bearing the extra burden of roadway deficits.

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Only Three States Raise Enough Dedicated Revenue to Fund Transportation Spending

State and Local Transportation Tax Revenue as Percentage of State Highway Spending, Fiscal Year 2021



Source: U.S. Census Bureau, "2021 State & Local Government Finances"; Federal Highway Administration, "Highway Statistics Series."

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Gas taxes are also regressive. Vehicles with lower fuel efficiency pay more per mile driven, subsidizing the vehicles with better fuel efficiency which are often newer and more expensive, even if these vehicles do no less damage to road surfaces. Electric vehicles don't consume gasoline at all. Rural drivers tend to drive lower gas mileage vehicles, so

they cause much more damage to the road than more fuel-efficient vehicles, like passenger cars, which cause less damage to the road.

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A VMT tax would provide a more stable source of funds to road spending and would not be subject to the same fluctuations as gas taxes. A VMT tax would also encourage more efficient driving and reduce the need for continual changes to fuel efficiencies, vehicle types, and consumer preferences.

Considering the Trade-Offs

Properly designing and implementing a VMT tax is not so easy, however. Policymakers must counterbalance simplicity with effectiveness, weighing the administrative costs of the system against the road use accuracy of a VMT measure. They must also consider the very legitimate privacy concerns the public has with government trying to track their driving and must design a system within the bounds of the Constitution (issues have been raised that could preclude mandatory GPS tracking as violative of the 4th, 9th, and 14th Amendments).

Low technology options for VMT monitoring include manual submissions of odometer readings (quarterly or annually as part of a state's existing inspection or registration process), time, and/or mileage permits. More advanced technologies include plug-in devices and smartphone apps, similar to those already available for use in insurance markets (e.g., Progressive's Snapshot and State Farm's Drive Safe & Save).

More accurate tracking of vehicles could be more expensive to implement and risk a greater invasion of privacy but would enable proper apportionment of funds based on the location of the driving. To help protect privacy, states operating VMTs rely on private-sector commercial account managers (CAMs) such as Azuga and Emovis. Data from individual users is sent directly to the CAMs. That data is then stripped of nonessential information and passed onto government agencies with only the information necessary for tax purposes (e.g., aggregate miles driven on public roads).

Once operational, VMTs have even broader applications. VMTs can vary road prices base pricing into the VMT tax. This d more complexity to the

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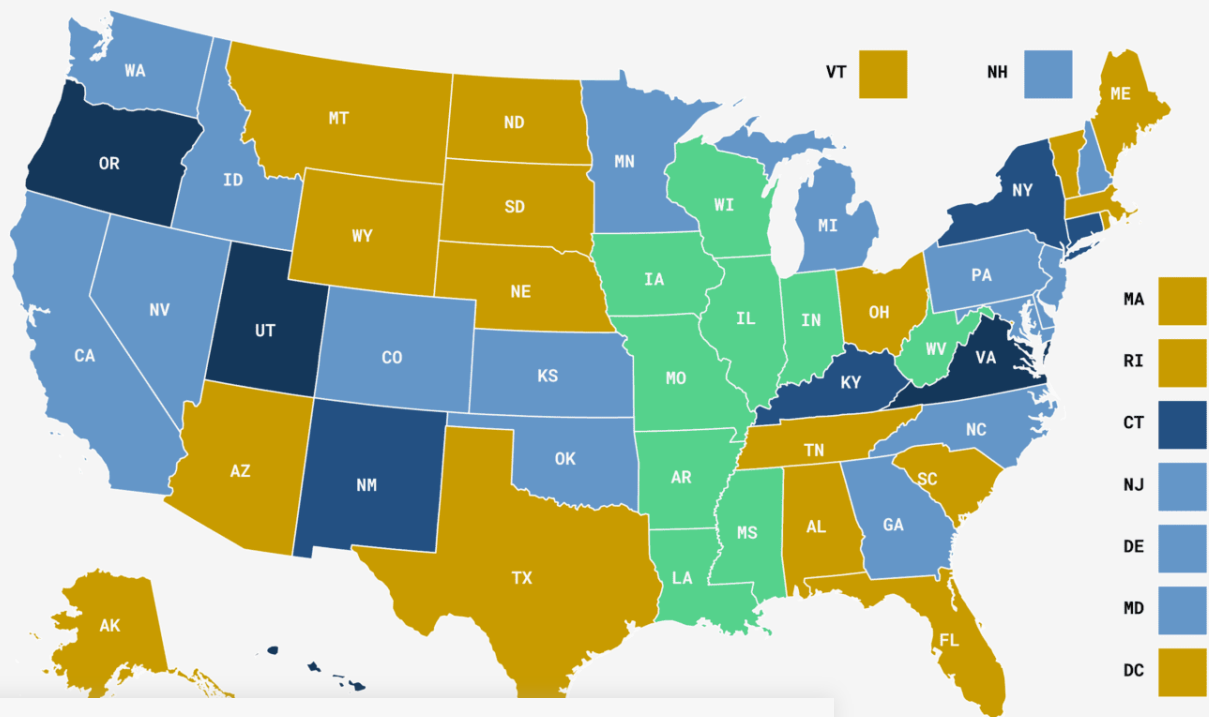
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her challenge for VMTs.
akers to work together to
research, design, and implement road user fees. The Eastern Transportation Coalition

brings together 20 eastern states on all manner of transportation projects, including mileage-based user fees, and Road Usage Charge America brings together 20 states specifically to research and advance road user fees.

Oregon was the first state to begin research into VMT taxes in 2001 and was the first to implement a program in 2015. Four states now have active programs for passenger vehicles and four other states have active programs targeting heavy commercial vehicles (Oregon has both), with pilot programs carried out in 16 states. Only Hawaii has a mandatory program, which requires EVs to participate by 2028 and all light vehicles by 2033.

Most States Are Taking Steps Toward VMT Taxes

Current Status of VMT Taxes across States, 2024



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■ State research &/or participation in regional coalition

No involvement with VMT

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Pricing varies by state. Drivers of a vehicle with an MPG greater than 20 can enroll in Oregon's OReGO system, which offers GPS or non-GPS tracking methods, credits for out-of-state driving using GPS, and prices per mile at \$0.02. Only EV drivers can enroll in Utah's Road Usage Charge (RUC) to pay \$0.016 per mile, tracked with an OBD device and a smartphone app—which includes location data but does not exempt out-of-state driving or private roads. Drivers of fuel-efficient vehicles in Virginia may enroll in the Mileage Choice Program to pay a variable rate based on MPG (up to \$0.0107 per mile for EVs), tracked with or without GPS but never with exemptions for private roads or out-of-state driving. Hawaiian EV drivers can enroll in their RUC next year, paying \$0.008 per mile as recorded by an annual odometer reading.

States have also tried various incentives for participation in these voluntary programs. Typically, VMT participants' charges are capped at the amount they would have been charged through the state's alternative system (e.g., Virginia's Highway Use Fee or alternative EV registration fees). This allows some participants—those that use the roads very little—to pay less in total taxes and fees. Also, VMT participants are able to spread their payment out over the entire year, as opposed to having to pay a one-time registration fee, and some participants, typically those in pilot programs, are given a small financial payment for participating.

A Principled Vehicle for Roadway Funding

The consensus from the assorted state studies, pilots, and programs is that VMT taxes work and they do not need to be overly complex or invasive. Highway deficits will continue to grow without reform to the gas tax. A VMT tax would establish the price of a mile,

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opportunities for “free-riders”—(out-of-state) users of the road who don’t pay into the state VMT. Patchwork regional coalitions and expanded toll systems can help bridge this gap to avoid a kind of transitional gains trap.

In time, VMT taxes can efficiently replace the existing structure; VMT taxes shouldn’t be seen as an add-on to the existing system. A VMT tax can be a simple, transparent, neutral, and stable “vehicle” for roadway funding.

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